

Project Planning

ComicCraft - AI Comic Story Creator using Gemini Models
Madras Institute of Technology | Department of Computer Technology (CT) | III Year, V Semester
Project period: 1 Oct 2026 - 2 Oct 2026

Project duration

Start date: 1 October 2026. End date: 2 October 2026. Total: 2 days.

Plan

Phase	Date	Work	Output	Status
1 Setup	1 Oct	Folder structure, venv, requirements, .env	Running FastAPI skeleton	Done
2 Text AI	1 Oct	Gemini outline and story, model configuration	generate_outline, generate_story	Done
3 Image AI	1 Oct	Stable Diffusion pipeline, MPS, warm-up	generate_image	Done
4 Layout and PDF	1 Oct	Layout builder, FPDF with DejaVu fonts	Preview and PDF	Done
5 UI	2 Oct	Home page redesign, background, pills	index.html	Done (preview/export pages basic)
6 Optimisation	2 Oct	fp16, parallel story, warm-up	~12.4 s per image measured	Done
7 Testing and docs	2 Oct	Smoke tests, end-to-end run, documents	Docs folder	Done

Sprints

Sprint	Dates	Goal
1	1 Oct	Backend, Gemini integration, image generation, layout and PDF export
2	2 Oct	UI, performance, quota and error handling, testing, documentation

Risks and mitigation

Risk	Mitigation
Gemini free-tier quota (20/day/model)	Use different models for outline and story; models set in .env
Model retirement	Model names configurable
Slow generation on CPU	Use GPU/MPS fp16; warm-up at start
Out of memory	Attention slicing, fp16, one image at a time
Leaked API key	Key only in .env, .gitignore